

Hyphae Module Analysis: BatchPrep Manager (BPM) - "The Kitchen Manager"

Date: November 27, 2025

Context: Food Service Operation (Single Operator Focus)

Module Goal: To serve as the "Second Brain" for the operator, managing cognitive load regarding production, safety, and inventory before service begins.

1. Core Philosophy: The "Push" Workflow

Most kitchen apps are "Pull" systems (the user opens the app to look for information). For a single operator, the **BatchPrep Manager (BPM)** must be a "Push" system. It should tell the operator what to do next based on inventory levels and sales forecasts, rather than waiting for the operator to decide.

The Golden Rule: *If it isn't prepped, it doesn't exist to the POS.*

2. Functional Requirements Analysis

A. Procurement & Dynamic Par Levels

The Problem: Buying is usually disconnected from prep. A single operator often forgets one crucial item (e.g., salt) that halts production.

- **Recipe Explosion:** When a production plan is set (e.g., "Make 500 Tacos this week"), the **Kitchen Manager** must explode recipes into constituent ingredients (Mise-en-place).
- **Inventory Delta:** Required Prep - Current Stock = Shopping List.
- **Vendor Grouping:** The list must auto-sort by vendor (e.g., "Produce Market," "Butcher," "Dry Goods").
- **Packaging as an Ingredient:** Packaging (boxes, liners, sauce cups) must be treated as Primary Ingredients. If you have 50kg of meat but 0 boxes, your sales capacity is 0.

B. The Production Logic (Multi-Focus Engine)

The Concept: Not all prep is equal. The system must distinguish between **Chemistry** (Baking), **Physics** (Sauces), and **Mechanics** (Knife Work) to manage the operator's attention effectively.

- **Type 1: The Baking Engine (Chemistry & Precision):**
 - *Focus:* High upfront precision, then Passive time.
 - **Baker's Math & Auto-Scaling:** If the operator inputs "850g flour available" (vs 1000g required), the BPM instantly recalculates water, yeast, and salt to maintain the exact hydration percentage.
- **Type 2: The Sauce Engine (Viscosity & Time):**
 - *Focus:* Intermittent attention (Stirring/Checking).

- **Multi-Stage Timers:** Standard timers fail here. The logic follows: *Stage 1 (High Heat)* -> Alert -> *Stage 2 (Simmer)* -> Alert -> *Stage 3 (Viscosity Check)*.
- **Cooling Logs:** Mandatory input of temperatures at specific intervals (HACCP) before storing.
- **Type 3: Fresh Produce & Raw Prep (Mechanics & Hygiene):**
 - **Focus: 100% Active Attention** (Knife work). This is the "Filler" activity during Passive baking/simmering times.
 - **The "No-Kill-Step" Protocol:** Unlike sauces or bread, raw salsa/onions have no heat kill step.
 - **Sanitization Check:** System mandates a "Sanitizer Soak Timer" (e.g., Microdyn for 10 mins) for lettuce/cilantro before chopping can begin.
 - **Oxidation Alerts:** For items like Avocados or Apples, the system flags urgency: *"Cover immediately. Shelf life < 4 hours if exposed."*
 - **Visual Consistency:** Displays photos of the required cut (e.g., *"Fine Dice 3mm"* vs *"Rough Chop"*) to ensure the taco topping is consistent every time.
 - **Allergen Interlock:** If the previous task involved nuts or shellfish, the system **locks** the screen and demands a "Wash/Sanitize Board" confirmation before unlocking the Fresh Produce module.

C. Inventory & Caducity (Shelf Life/Expiration)

The Problem: "First In, First Out" (FIFO) is hard to track mentally when you have multiple batches of the same sauce.

- **Batch ID Generation:** Every completed prep task generates a unique Batch ID (e.g., SAUCE-ROJA-1127).
- **Auto-Caducity Calculation:**
 - **Logic:** Prep Date + Defined Shelf Life = Expiry Date.
 - **Example:** Salsa Verde made on Nov 27th with a 3-day life expires Nov 30th.
- **Label Printing:** The touchscreen must interface with a label printer. One tap prints: *Item Name, Prep Date, Expiry Date, Batch ID, Operator Name, and Major Allergens.*
- **Visual Decay (Accessible Design):** The BPM dashboard must use **Double Coding** (Color + Icon) to ensure accessibility for operators with CVD (Color Vision Deficiency):
 - **Fresh:** Green/Blue + **Checkmark Icon.**
 - **Use Soon:** Orange/Yellow + **Warning Triangle Icon.**
 - **Expired:** Red/Vermilion + **'X' or Trash Can Icon.** (Avoid using Red/Green alone as a differentiator).

D. Costing & Waste (The Silent Killer)

For a single operator, waste is a direct salary deduction.

- **Theoretical vs. Actual Cost:**
 - **Theoretical:** Recipe says 100g cheese per taco.
 - **Actual:** Operator uses 110g or drops 500g on the floor.
- **The "Spill" Button:** A quick-access button to log waste immediately during prep.

"Dropped," "Spoiled," or "Burned."

- **Yield Tracking:**

- Recipe expects 2 Liters yield.
- Operator logs actual yield of 1.8 Liters.
- System adjusts the *Cost Per Unit* instantly based on the actual yield, not the theoretical yield.

E. Product Recycling & Upcycling (The Circular Kitchen)

The Concept: Converting "Potential Waste" into "Active Inventory." The BPM must handle recursive recipes where a finished product becomes an ingredient for a new product.

- **"Convert" Action:**

- Instead of marking an item as "Expired/Waste," the operator selects **"Convert"**.
- *Example:* "Hamburger Buns (Batch A)" -> Convert -> "Stale Bread (Ingredient)".
- *Example:* "Sweet Potato Fries (Batch B)" -> Convert -> "Sweet Potato Hash Base (Ingredient)".

- **Cost Basis Transfer:**

- When converting, the cost of the old item transfers to the new item.
- *Logic:* The Bread Pudding cost includes the original cost of the Buns, ensuring profit margins remain accurate even on recycled goods.

- **Safety & Caducity Reset (The Kill Step):**

- The system must distinguish between *Repackaging* (Date stays same) vs. *Re-cooking* (Date resets).
- If the conversion involves a "Kill Step" (cooking above 165°F/74°C), the Expiry Clock resets.
- *Traceability:* The new "Bread Pudding" Batch ID must link back to the original "Bun" Batch ID in the database for safety recall purposes.

F. Hybrid Production Models (Service vs. Retail)

The Concept: The ability to handle "One Batch, Two Destinations." A food cart operator often makes a large batch of sauce and wants to sell 80% on tacos and jar 20% for retail sale.

- **The Split-Yield Workflow:**

- When completing a batch (e.g., Pickled Chiles), the **Kitchen Manager** should ask: **"Destination?"**
- *Option:* **Split Batch.**
- *Input:* "3 Liters to Service Line (Cambro)" AND "10 Jars to Retail (8oz Jars)."

- **Retail Packaging Logic:**

- If "Retail" is selected, the system *must* check stock for specific retail packaging (Jars, Lids, Safety Seals, Brand Labels).
- If jars are out of stock, the system warns the operator *before* they start cooking.

- **Dual Labeling:**

- **Service Label:** Simple, functional (Date, Name, Expiry).
- **Retail Label:** If the printer supports it, this outputs a consumer-facing label

(Ingredients, Net Wt, Barcode) or prompts the user to apply pre-printed brand stickers.

G. Operations & Hygiene (Opening/Closing Logic)

The Concept: The "Digital Logbook." It bridges the gap between shifts, ensuring the kitchen isn't just "productive" but also "reset" correctly. This replaces paper checklists.

- **Time Logging (The Shift Clock):**
 - **Philosophy:** "Self-Control," not "Big Brother." The goal is data, not surveillance.
 - **Action:** Simple "Check-in" button on startup and "Check-out" on shutdown.
 - **Value:** Allows the HCA to calculate the true **Labor Cost** of the session. (e.g., "This batch of Tamales took 4 hours of labor"). This helps the operator realize if they are charging enough for their time.
- **Opening Protocol (The "State of the Union"):**
 - **Prior Shift Carryover:** The first screen upon login *must* show notes/flags from the previous closing. (e.g., "Leftover sanitizer bucket," "Slicer blade needs sharpening").
 - **CCP Verification:** Mandatory prompt to log critical metrics before unlocking the "Prep" module (e.g., Fridge Temps, Pest Checks).
 - **"Lazy Closer" Detection:** If a closing task was marked "Skipped" the night before, it appears as a High Priority "Must Do" task at opening.
- **Closing Protocol (The "Reset"):**
 - **The Zero-Out List:** A mandatory checklist to power down (Trash Out, Gas Off, Surfaces Sanitized).
 - **Deep Clean Rotation:** The system tracks non-daily tasks. "Clean Hood Filters" or "Defrost Deep Freeze" appears automatically on Sundays or the 1st of the month.
 - **Inventory Reconciliation:** A quick "Eye-Ball" check. "Do we actually have milk for tomorrow?" (Prevents morning surprises).
- **Hygiene Accountability:**
 - **User Attribution:** Every "Cleaned" checkbox is stamped with the User ID and Timestamp.
 - **Photo Verification (Optional):** For critical items (like grease traps or night locks), the app can request a photo proof before allowing the "Close Day" action to complete.

H. Multi-Tasking Engine (The "Tetris" Schedule)

The Concept: A single operator is a bottleneck. To maximize output, the system must schedule **Active** tasks during the "downtime" of **Passive** tasks.

- **Active vs. Passive Logic:**
 - Every recipe step is tagged:
 - **Active:** Requires Hands/Eyes (Chopping, Whisking, Weighing).
 - **Passive:** Requires Time/Equipment (Roasting, Proofing, Simmering, Cooling).
 - *The Algorithm:* The BPM identifies the longest Passive task (e.g., "Proof Dough: 2 hours") and schedules Active tasks (e.g., "Dice Onions") to fill that gap.

- **Resource Management (Hardware Slots):**
 - The system tracks constraints to prevent "Double Booking" equipment.
 - *Logic:* If "Bake Buns" requires Oven_Slot_1 and "Roast Tomatoes" requires Oven_Slot_1, the system schedules them sequentially. However, "Boil Salsa" (Burner_Slot_1) can be scheduled parallel to the Oven task.
- **The "Context Switch" Alert:**
 - 5 minutes before a Passive task finishes (e.g., Oven Timer ending), the app alerts the operator: *"Wrap up onion dicing. Buns need rotation in 5 mins."* This prevents burnt food caused by hyper-focus on a cutting task.

I. Goal-Oriented Production (The "North Star")

The Concept: Moving from "Maintenance Mode" (filling pans) to "Mission Mode" (achieving a sales target). This uses **Backward Chaining** logic.

- **Dependency Tree (Parents & Children):**
 - *Input:* "Goal: Sell 200 Burgers this Saturday."
 - *Explosion:* The system calculates the Bill of Materials:
 - **Parent:** 200 Burgers.
 - **Child A:** 200 Buns (5 Baking Cycles).
 - **Child B:** 6L Sauce (1 Batch).
 - **Child C:** 30kg Meat Mix (Active Grinding).
 - *Gap Analysis:* The system checks current inventory ("You have 50 buns") and adjusts the goal ("Make 150 buns").
- **Visual "Batch Bricks" (Gamification):**
 - Large goals are broken into manageable "Bricks" (Batches).
 - *Visual:* A stack of 5 bricks for the Buns.
 - *Status Colors:* Gray (Planned) -> Pulsing Blue (In Oven) -> Green/Checkmark (Done).
 - *The Psychology:* The operator's job is to "Turn the pyramid Green."
- **The Caducity Timeline:**
 - The system schedules production based on shelf life, not just availability.
 - *Monday:* Schedule High-Shelf-Life items (Sauces, Frozen Buns).
 - *Friday:* Schedule Low-Shelf-Life items (Fresh Veg, Thawing Meat).

J. Logistics & Load-Out (The Kitchen-to-Cart Handshake)

The Concept: The critical workflow of moving stock from the "Prep Kitchen" (Fixed) to the "Cart/POS" (Mobile). This ensures the cart never leaves the kitchen missing a vital ingredient.

- **The "Load-Out Manifest":**
 - Derived directly from the **Goal**. If the goal is "200 Burgers," the Load-Out Manifest automatically lists: *200 Buns, 200 Patties, 6L Sauce, 200 Napkins, 1 Full Propane Tank*.
 - *Logic:* It acts as a "Pick List" for the operator to physically load the cart.
- **The Digital Handshake (Inventory Transfer):**
 - **Action:** Operator taps "Transfer to Cart" on the Kitchen Manager.

- **Handshake:** The data is pushed to the HCA (Backend), which then updates the POS "Opening Stock."
- **Result:** When the operator opens the POS on the cart at the event, the inventory is *already populated*. No manual counting required on the street.
- **The "Pre-Flight" Safety Check:**
 - Before the "Transfer" is finalized, the system demands a "Pre-Flight" verification:
 - *Temps:* "Is the cart fridge at 3°C?"
 - *Power:* "Are POS batteries/Power banks charged?"
 - *Consumables:* "Paper rolls? Sanitizer spray?"

K. Ingestion & Receiving (The Firewall)

The Concept: Establishing a "Firewall" at the kitchen entry point. This manages inventory and cost updates for bulk purchases where formal receipts are often absent or simplified (e.g., purchases from Central de Abastos).

- **The "Receiving Mode" Workflow (Simplified):**
 - **Action:** Operator taps "Start Receiving." The app displays the items from the **Shopping List** generated earlier.
 - **No Invoice Matching:** The focus is on capturing the three critical variables instantly: Quantity, Unit, and Price.
 - **Unit Conversion (Bulk to Standard):** When receiving a bulk item (e.g., "Onions"), the app prompts for the physical unit received and its net weight/count.
 - *Input:* "Received: **1 Arpilla** (Sack)." -> Prompt: "Net Weight (KG)?" -> Operator inputs 25.
 - *System Logic:* Updates the HCA with "1 Arpilla = 25kg."
 - **Manual Price Confirmation (Cost Averaging):** Instead of scanning an invoice, the operator inputs the total cost for the item.
 - *Input:* "Total Cost for 25kg Onions?" -> Operator inputs \$300.
 - *Result:* The BPM pushes the new **Cost Per Unit** (\$12/kg) to the HCA, instantly recalculating the average cost of the recipe.
- **Raw Goods Labeling (FIFO Start):**
 - After the item is accepted, the label printer is triggered.
 - *Label:* "RAW - Onion - Rec'd Nov 27 - Expire Dec 10." (Mandatory label for every new incoming item to ensure FIFO compliance).

3. UX/UI Analysis: The "Hostile Environment" Design

The kitchen is a hostile environment for electronics (heat, grease, water, flour, noise).

A. Touchscreen Ergonomics

- **Target Size:** Buttons must be minimum 60px x 60px.
- **Gestures > Taps:** Swipe right to "Complete," Swipe left to "Discard."
- **High Contrast Mode:** Dark mode with high-luminance colors (Cyan, Magenta, Bright

Yellow) for legibility.

B. The "Flight Control" Dashboard (Parallel View)

- **Zone 1: The Focus Zone (Center):** Displays the single **Active** task currently requiring hands (e.g., "Whisking"). Large text, step-by-step.
 - **Container Guide:** Explicitly tells user: "Use 4L Cambro (Green Lid)." Eliminates guessing.
- **Zone 2: The Monitor Zone (Sidebar):** Tracks running **Passive** tasks.
 - *Visual:* Small cards with progress bars. [Icon: Oven] Buns: 15m left.
 - *Interaction:* Tapping a card swaps it to the Focus Zone.

C. Audio Feedback (No Voice in v1)

- **Audible Alarms:** Distinct sounds for "Timer Done" vs. "Critical Alert."

D. Navigation & Input Strategy (The "No-Keyboard" Rule)

- **Zero-Scroll Architecture:** Fixed-height cards with "Next" arrows.
- **Simplified Inputs:** Contextual numpads, Smart Presets ([1kg] [5kg]), Steppers (+ / -), and Barcode scanning.
- **Task Acceleration (The Power User Feature):**
 - **Group Completion (Macro-Tasks):** Allows the operator to tap and select several sequential, low-stakes steps (e.g., "Measure Salt," "Measure Yeast," "Measure Sugar") and mark them **Done** simultaneously with a single confirmation tap.
 - **Safety Gate:** The system must only allow group completion for steps that do NOT involve HACCP logging (temperature/pH), high waste potential, or the Allergen Interlock. If any of those are present in the skipped steps, the system forces manual confirmation.
 - **Focus Zone Jump:** A single, large "Skip to Next Critical Step" button instantly bypasses all rote steps (e.g., "Grab Container," "Set Aside") to move immediately to the next required action (e.g., "Start Mixer").

4. Typical Workflow Scenario (The "Goal-Driven Day")

1. **08:00 - The Mission:** Operator views the "Goal Pyramid." Target: "200 Burgers for Saturday."
2. **08:05 - The Strategy:** BPM suggests: "Start Dough (Active) now. While it proofs (Passive), make Salsa."
3. **08:15 - Parallel Prep:**
 - **Focus:** Operator is kneading dough.
 - **Monitor:** Empty.
4. **08:45 - The Switch:**
 - Dough goes into container. Timer set: "Proof 2 Hours" (Passive).
 - BPM prompts: "Dough is rising. Switch to Salsa Prep?"
5. **10:45 - The Alert:**

- Salsa is simmering (Passive). Dough timer beeps: "Proof Complete."
 - BPM guides operator to "Focus Zone: Shape Buns."
6. **12:00 - Progress:** Operator taps 2 "Batch Bricks" to Green. The Pyramid fills up.

5. Integration Architecture: The HCA Link

- **Role of the HCA:** Single Source of Truth, Forecasting Engine, Global Cost Updates.
- **Data Synchronization:** BPM pushes inventory/waste logs; HCA pushes production orders.
- **Offline Capability:** BPM caches HCA data for local execution.

6. Future Roadmap (Version 2.x)

- **Voice Activation & Control:** Hands-free navigation and logging.
- **AI-Vision Verification:** Camera-based mise-en-place checks.

7. Development Strategy: Data & Prompts

Part A: Data Architecture Concepts (Database Implications)

To support the complexity of the "Tetris" scheduler and the "Goal-Driven" production, the database schema must be robust. These concepts should feed directly into the development prompts.

1. **items (Global):** Raw ingredients, packaging, and sub-recipes.
2. **recipes:**
 - `step_type`: "Active" or "Passive".
 - `resource_required`: "Oven", "Burner", "Mixer".
 - `batch_size`: For "Brick" calculation.
3. **production_goals:** Links a "Sales Event" to multiple "Batch" requirements.
4. **batches:** An *instance* of a recipe. Contains `created_at`, `expires_at`, `yield_actual`.
5. **resources:** Available hardware slots (e.g., `oven_capacity`: 2).
6. **audit_logs:** For safety. Who made it? What temperature was it cooled to?
7. **suppliers:** Linked to items for the automated shopping list.

Part B: Development Prompts (Building the App)

The Concept: A collection of highly specific prompts designed to be fed into AI development tools (like Mocha, Base 44, or Cursor) to generate the actual code structure, UI components, and logic for the BatchPrep Manager.

- **Prompt Set 1: Generating the "Tetris" Scheduling Logic**
 - *Objective:* Create the algorithm that interleaves Active and Passive tasks.
 - *Prompt:* "Create a TypeScript class named `ProductionScheduler`. It should accept an array of Recipe objects. Each recipe has steps with a type ('active' or 'passive') and duration. The scheduler must identify the longest 'passive' step (e.g., 'Proof Dough: 60min') and fill that time slot with 'active' steps from other recipes (e.g., 'Chop

Onions: 20min'). Return a timeline object organized by time blocks. Utilize the resources table structure to check for equipment conflicts."

- **Prompt Set 2: Generating the "Flight Control" UI**

- *Objective:* Build the split-screen dashboard for the kitchen tablet.
- *Prompt:* "Generate a React component using Tailwind CSS for a kitchen display system. The layout must be a split-screen. **Left Side (60%):** A 'Focus Zone' displaying a single instruction step in massive text (text-6xl), with a 'Done' button that is 100px high. **Right Side (30%):** A 'Monitor Zone' displaying a vertical list of running timers (e.g., 'Oven 1: 15m remaining'). Use a dark theme (bg-slate-900) with high-contrast text."

- **Prompt Set 3: Generating the "Ingestion/Receiving" Module**

- *Objective:* Handle bulk unit conversions and cost averaging.
- *Prompt:* "Create a data model and input form for 'Bulk Receiving'. The form needs a toggle for 'Unit Type' (Standard vs. Bulk). If 'Bulk' is selected (e.g., 'Crate'), dynamically appear a field for 'Net Weight'. Include a field for 'Total Cost Paid'. Write a function `calculateCostPerGram(totalCost, netWeight)` that updates the `inventory_item` record."

- **Prompt Set 4: Generating the "Goal-to-Prep" Explosion**

- *Objective:* The Backward Chaining logic.
- *Prompt:* "Write a backend function `explodeProductionGoal`. Input: `goal_id` (e.g., 200 Burgers). Logic: Query the Recipe table recursively to find all child ingredients (Buns, Patties, Sauce) and their quantities. Check against the Inventory table. Output a JSON object listing: `items_to_prep`, `batches_required`, and `shopping_list_items`."

- **Prompt Set 5: Generating the "Shift Clock"**

- *Objective:* Simple time tracking without surveillance vibes.
- *Prompt:* "Create a minimal 'KitchenShift' component. It needs one large button that toggles between 'Check In' (Green) and 'Check Out' (Red). Store the timestamps in a `ShiftLogs` collection. Display a running 'Session Duration' timer that is visible but unobtrusive (small font, top corner)."